

Business Themes in the Trademark Record: Language Signals of Product Survival, Funding, and Listing

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Abstract

Trademark goods/services descriptions are a forced, dated disclosure of what a firm expected to sell, written by the millions of firms that never patent. This paper builds two measures on them and tests what each predicts. A description is reduced to themes by a topic model fitted once over the corpus; its theme mix is compared, by Kullback–Leibler divergence, to its own Nice class in the five years before and after filing. The average of the two divergences is atypicality; their difference is lead. Three partitions of the vocabulary (fifty global themes, five hundred, fifty per class), two reference weightings, and five window constructions are compared on identical filings. Lead is robust to all of them and predicts cancellation for non-use at the five-year proof of continued use: +2.3, +2.6 and +1.9 points under the three partitions, +1.4 with design controls ($t = 8.3$). Atypicality is not: its protective effect at the proof is 4.4 points under global themes and 0.7 under per-class themes, and its sign at the listing margin depends on how description length is held. Two hazards are documented for trademark-text research. Scoring words rather than theme distributions measures the length of the filing; and the registered description is a conformed document, because applicants who refile rewrite toward their class’s typical language. The measure’s validity is therefore strongest for lead, for outcomes at the mark, and within an industry; the paper states where each condition fails.

1 Introduction

The pioneers are the ones with arrows in their backs. The claim has been tested on named markets and naming choices (Golder & Tellis, 1993; Semadeni & Anderson, 2010), but not on a record where being early can be read directly from what every entrant wrote, because that record did not exist in usable form. It does now. Every US trademark application since 1884 is public, with its goods/services description, its owner, its counsel of record, and a dated prosecution history that records whether the mark registered, whether its owner proved five years later that the product was still in commerce, and whether it was renewed at ten. This paper re-parses all 13.99 million case files into that dated form and asks of each filing two questions about its language: how unusual was this description for its industry, and was it written in words the industry was moving toward or words it was leaving behind. The first is the filing’s *atypicality*, the second its *lead*. The paper then follows the filing, and the firm behind it, through the gates the record provides.

Section 2 places the exercise: in the trademarks-as-data literature, which counts filings but has not read their text against time; in the text-measurement literature the construction descends from; and in the pioneering and optimal-distinctiveness literatures whose question it answers at census scale — including the nomenclature, since several established terms assert mechanisms this design cannot show. Section 3 builds the measure: a description is reduced to the themes it draws on,

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fitted once over the corpus by a topic model; its theme proportions are compared, by Kullback–Leibler divergence, to the average proportions of its own Nice class in the five years before its filing date and the five years after. The average of the two divergences is atypicality; their difference is lead, written L . The section states why words are not scored directly, why surprise is measured within a class and never across classes, and the record’s gates and counting conventions.

Sections 4–7 carry the results. Registration rises with atypicality and turns down only in the most atypical decile, is nearly flat in lead — a quantity no examiner could observe, since it is computed from filings not yet made — and applicants who refile an abandoned application rewrite toward the class’s typical description, so every registered description is in part a conformed one (Section 4). The central result is the five-year proof of continued use: among 3.10 million registrations, marks whose language most led their industry were cancelled 1.4 points more often than marks whose language most lagged it, within Nice class and registration year with drafting and counsel held — 2.3 points raw, in 33 of 44 classes, robust to the timing of failure, to who files, to atypicality strata, to text-duplication, and to three partitions of the vocabulary; atypical marks survive the proof more often, but that protection belongs mostly to filings extending a theme established elsewhere into a new class (Section 5). The penalty is episodic where language moved fastest: two to three times larger for marks filed in the late 1990s, reversed for 2000–2004 filings, and returned from 2008, with the swing inside four technology classes and, within them, in which themes each era’s leading filings were drawn from (Section 6). Following the owner instead of the mark, the debut’s language does not predict a priced private round; SEC reporting and listing fall in lead; and the firms that came to hold most of the linked value described their first product in language their industry already used (Section 7).

Section 8 reads the stages together on one harmonized sample and states the implications for trademark-based measurement. Section 9 reports what changes under other choices of resolution, weighting, window, and period; one choice, the decayed reference window, shifts the headline estimate, by about a seventh. The full working-paper treatment, an interactive per-class appendix, and every artifact behind every number are public.

Being early with the language of a product costs at the five-year proof of continued use, in three of every four product classes and under every partition of the vocabulary, and it costs most in the industries and the years in which the language was moving fastest. Being unusual is rewarded at the same proof, mostly where a theme is imported from elsewhere, and it is the examiner, not the market, that rewards middling conformity at registration. The firms that thrived were not the pioneers of language. They were written in the words their industries already used, and they are few.

2 Related literature and nomenclature

2.1 Trademarks as innovation data

A trademark filing is a dated, costly, public disclosure made by the millions of firms that never patent, and a literature has grown around reading it as an innovation indicator (Castaldi, 2020; Willeke et al., 2026). Trademark counts and breadth predict venture valuations (Block et al., 2014); filing timing tracks innovation processes (Seip et al., 2018); and trademarks mark the product-market side of innovation that patents miss (Flikkema et al., 2019, 2025). Census-linked work reads registration as a correlate of firm growth and survival (Dinlersoz et al., 2018, 2023). Nearly all of it counts filings, classes, or renewals; the descriptions’ text enters, if at all, as breadth or similarity. This paper reads the text itself, and reads it against time.

2.2 Text measures of novelty

Scoring a document by its divergence from a reference corpus before and after it comes from cultural analytics: Murdock et al. (2017) on Darwin’s reading notebooks, Barron et al. (2018) on the French Revolution’s parliamentary debates. In economics and finance, text-based measures locate firms in product space (Hoberg & Phillips, 2016), date technological innovation in patents over the long run (Kelly et al., 2021), and track the vintage of patent language (Kalyani, 2024). The construction here differs in its object — the composition of commercial offerings, not ideas or claims — and in its two-sided reference: every filing is scored against its own industry’s language both before and after its date, which is what lets *unusual* be separated from *early*. The separation matters because the two components predict opposite things.

2.3 Pioneering and optimal distinctiveness

Whether market pioneers are advantaged or burdened is an old, unresolved question measured mostly on named markets and survivor-prone samples (Golder & Tellis, 1993; Semadeni & Anderson, 2010). A parallel literature finds evaluators reward conformity with a penalty for extremity in both directions — the optimal distinctiveness of Zuckerman (1999), Hsu (2006), Zhao et al. (2017), and Taeuscher & Rothe (2021). This paper meets both at census scale: every entrant’s language is observed, whether or not the product survived to be remembered, and the record supplies two different evaluators — an examiner at registration and the market at the five-year proof of continued use — who turn out to price the two components differently. The examiner rewards middling conformity and cannot observe lead; the market is indifferent to conformity once theme importation is held, and penalizes having been early.

2.4 Nomenclature

The innovation literature is not consistent about what quantities like these are called, and several of its established terms assert things this design cannot show, so the paper’s names are chosen against the nearest occupants rather than borrowed from them. Table 1 maps the five quantities; the entries below say, for each declined term, what adopting it would claim.

Table 1: The five quantities, their nearest occupants in the literature, and the terms used here. P is a filing’s theme distribution; Q_{past} and Q_{future} aggregate the five years of same-class filings before and after it (Section 3).

Quantity	Nearest in the literature	Term used here
$K^- = \text{KL}(P Q_{\text{past}})$	novelty (Barron et al., 2018; Murdock et al., 2017); originality (Hall et al., 2001)	past-facing surprise
$K^+ = \text{KL}(P Q_{\text{future}})$	transience (Murdock et al., 2017); generality (Hall et al., 2001), which measures realized influence, not position	future-facing surprise
$A = \frac{1}{2}(K^- + K^+)$	atypicality (Uzzi et al., 2013), which scores internal pairings, not distance from the class	atypicality
$L = K^- - K^+$	resonance (Barron et al., 2018), which asserts a mechanism	lead (leading / lagging)
$ L $	no established occupant	lead magnitude

Novelty and *transience* are the two KL levels themselves; novelty is declined as a name for anything here because it names only the backward-looking half of a quantity that faces both ways. *Resonance* is arithmetically what this paper calls lead, and is avoided because it names a mechanism — the field resonating with an author’s contribution — that this design cannot observe: a filing can sit where its class was heading because it was imitated, because both responded to the same outside event, or by coincidence.

Radicalness is the closest neighbour among the innovation labels, since KL is a departure measure rather than a newness count. It is declined because it claims existing competences are devalued, which nothing here licenses — and because the label itself is unsettled: a bibliometric review of 4,407 articles finds *radical*, *breakthrough*, *discontinuous* and *disruptive* used with definitions inconsistent within labels and not distinctive across them (Verhoeven et al., 2025).

Atypicality in the bibliometric sense (Uzzi et al., 2013), applied to patents as unconventionality by Berkes & Gaetani (2021), scores the rarity of the pairwise combinations inside a document against the population’s co-occurrence rates. The atypicality here is a different object: the distance of one document’s composition from its class’s, not the rarity of its internal pairings; the two are close to independent on this corpus (Section 3). The word is kept because both mean unlike the population; which population, and along what, differ.

Originality and *generality* (Hall et al., 2001) are the patent literature’s standard backward- and forward-facing pair, and generality is the nearest established forward-facing quantity to K^+ . Both measure the dispersion of citations across technology classes, and generality requires the forward citations to have arrived, so it observes realized influence; K^+ observes the class’s movement whether or not the filing influenced it.

Distinctiveness has two prior occupants: trademark law’s registrability doctrine, and strategic management’s optimal distinctiveness (Zhao et al., 2017), which predicts an inverted-U that appears here at registration but not at listing, so the term would imply a theory is being tested when it is not. *Innovation* is a property of the product, not of the sentence describing it. What *atypicality* says is narrower than any of these: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

3 The measure, the record, and the conventions

3.1 The record and the question

A US trademark application carries a description of the goods or services the mark will cover. The description is written to secure scope of protection: wide enough to cover the planned use, narrow enough to be granted. It is therefore a forced, dated disclosure of what a firm expected to sell. The USPTO publishes every application since 1884 with its prosecution history, and this paper re-parses all 13.99 million case files into dated events, so that each application can be followed from filing to registration, to the sworn proof of continued use required between the fifth and sixth year, and to cancellation or renewal.

The question is whether the language of the description, read against the language of the rest of the industry at the time, predicts what happened to the mark and to the firm behind it. Two properties of the language are measured. *Atypicality* is how unusual the description is for its industry. *Lead* is whether its language is ahead of or behind where the industry’s language was moving. Sections 4–7 relate the two to reaching registration, to the five-year proof of continued use, and to venture funding and public listing. Section 9 reports what changes under other choices.

3.2 Two readers, one living backwards

Imagine two readers of a filing’s goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other, living backwards as Merlin was said to, has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry’s language moved that way.

Both windows are anchored on the filing’s own date, spanning the 1,826 days before it and the 1,826 days after; both exclude the filing date itself, so a filing never appears in the reference of another filing made the same day, and no two filings made on different days share a reference. Inside a window every day counts the same. The measure asks whether wording is unusual against a period, not whether it is ahead of a trend inside that period; it has no momentum channel.

3.3 What is measured

The quantity is information content in Shannon’s sense. Something that happens with probability p carries $-\log p$ of information when it happens: a common thing carries little, a rare thing a lot. Here the things are the themes a description draws on, and the probabilities are how often the filing’s own industry drew on each of them in the reference window. A filing’s score is the average information content of its themes under its industry’s frequencies, weighted by how heavily it uses each, less the same average taken under the filing’s own frequencies.

The apparatus is that of Murdock et al. (2017), developed on Darwin’s reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates. They score a document against those before it and again against those after, calling the two quantities *novelty* and *transience* and their signed difference *resonance*. Their object was the life of ideas: how often an idea appeared, and how ideas stood in relation to one another, across one reading life or one assembly’s debate. The object here is the composition of commercial offerings, which components a product or service is described as having, and how that composition shifts across an industry over time. The arithmetic carries over unchanged. The interpretation does not: their document has one author and the reference stream is that author’s own reading, so a high value says something about a mind; here a filing is scored against other firms’ filings, and its author is usually counsel drafting for scope of protection.

Themes, not topics. The clusters the model fits are called *themes* throughout. In the source literature and in the name of the method, latent Dirichlet allocation, the same objects are called topics, and a document’s topic is what it is about. A goods/services description is not about anything in that sense; it lists components. A cluster of terms that co-occur across descriptions (*video, music, games, electronic, audio*) is a component that many offerings share, not a subject. Theme is the word for that. Topic is used here only in the name of the model.

3.4 From a description to two numbers

Counting words. A description is reduced to a bag of words: the text is lowercased, split into tokens of three or more letters, and counted, with adjacent pairs counted as tokens in their own right so that *computer software* registers alongside *computer* and *software*. Order is discarded. Take the filing for AMAZON.COM, lodged in the advertising and retail class on 18 April 1997:

computerized on line search and ordering service featuring the wholesale and retail distribution of books, music, motion pictures, multimedia products and computer software in the form of

printed books, audiocassettes, videocassettes, compact disks, floppy disks, CD ROMs, and direct digital transmission

Of its 45 distinct terms, 36 are in the theme model’s vocabulary, among them *computerized*, *search*, *ordering*, *ordering service*, *retail*, *distribution*, *printed books* and *digital transmission*.

Grouping words into themes. Each filing is then re-described in a fixed, much smaller set of coordinates. Latent Dirichlet allocation is run once over a stratified sample of the whole corpus, all 45 classes together: it learns 50 clusters of words that tend to occur together and, for any filing, the proportions in which its words draw on them. Nothing supervises this; the clusters come from co-occurrence alone. This build — fifty global themes, per-filing five-year windows, flat weighting — is the production scoring, used for every estimate unless the text states another. The themes are shared across classes; what is specific to a class is the reference, the average theme proportions of that class’s own filings in the window. The Amazon filing draws on six themes at more than a twentieth of its weight: 0.41 on media and entertainment goods (*video*, *computer*, *electronic*, *music*, *games*, *audio*), 0.15 on computer and information services (*services*, *computer*, *software*, *information*, *data*), 0.13 on retail store services (*store*, *retail*, *store services*, *featuring*), 0.10 on printed matter (*paper*, *books*, *printed*, *cards*, *stationery*), 0.09 on downloadable and online software, and 0.07 on marketing of consumer goods. The online appendix lists every theme with its leading words and its weight in each class.¹

Fifty is coarse, roughly one theme per industry. Section 5.3 refits the model at 500 themes and separately at 50 themes within each class and repeats the five-year-proof analysis under each; Section 9 reports the resolution sweep and a second fit at the same resolution with a different seed. Fifty is kept elsewhere because it is the coarsest resolution at which every result holds, and because its software-class lead contrast sits at the low end of the resolutions tried (only the 100-theme refit returns less).

Comparing against the two references. Write P for the filing’s distribution over themes, and Q_{past} and Q_{future} for that distribution averaged over every same-class filing in the two windows, $[d_i - W, d_i]$ and $(d_i, d_i + W]$, with $W = 1,826$ days. How far P sits from a reference Q is the Kullback–Leibler divergence,

$$\text{KL}(P \parallel Q) = \sum_k P_k \log \frac{P_k}{Q_k},$$

a sum over the 50 themes, zero when the two distributions agree and growing as the filing puts weight where the industry puts little. It is measured in nats, the natural-log unit. Write

$$K^- = \text{KL}(P \parallel Q_{\text{past}}), \quad K^+ = \text{KL}(P \parallel Q_{\text{future}}),$$

so K^- is what the first reader feels and K^+ the second: the filing’s past-facing and future-facing surprise.

Taking the difference. Amazon’s filing is scored against 41,166 earlier filings and 146,532 later ones, giving $K^- = 1.286$ and $K^+ = 1.163$. The two axes used throughout are their average and their signed difference:

$$\underbrace{A = \frac{1}{2} (K^- + K^+)}_{\text{atypicality}}, \quad \underbrace{L = K^- - K^+}_{\text{lead}}.$$

¹https://aporia.institute/tm-vocabulary/online-appendix/themes_T50.html

For Amazon, $A = 1.225$ and $L = +0.123$: against its own class, its atypicality is at the 47th percentile and its lead at the 94th. The industry moved toward this description in the five years after it was filed and had not been writing that way in the five years before. Table 2 maps the terms; the signed difference is written L throughout and its magnitude $|L|$.

What “early” means is visible in the themes themselves: the three themes that carry most of Amazon’s description — retail store services, media and entertainment goods, and the still-small online-software theme — all became more common in class 035 over the following five years, the last nearly doubling, while the two minor themes it touched receded. The filing is closer to the class’s future composition than to its past one; the supplementary material tabulates the six theme shares.

K^- and K^+ correlate at 0.988 on the software class, so entering both is close to entering one variable twice; rotating them into an average and a difference isolates the large common component from the small residual that carries the timing, and leaves the two regressors nearly uncorrelated ($+0.036$ against each other, where K^- alone would give $+0.114$). The cost is that L is a small residual of two large quantities, about a sixth of either level’s spread, which Section 9 quantifies.

Table 2: The quantities, their sources, and the terms used here. *Lagging* and *leading* describe the two signs of L ; *atypicality* is unsigned and says nothing about direction.

Quantity	Definition	Murdock et al.	Term used here
Surprise against the past	K^-	novelty	past-facing surprise
Surprise against the future	K^+	transience	future-facing surprise
Their average	$A = \frac{1}{2}(K^- + K^+)$	—	atypicality
Their signed difference	$L = K^- - K^+$	resonance	lead (leading / lagging)
Its magnitude	$ L $	—	lead magnitude

Why the words are not scored directly. The themes are an extra step, and the words themselves could be compared to the reference: build the filing’s distribution over the class vocabulary from the words it uses, do the same for every filing in the window, and take the divergence. The paper reports that scoring as a cross-check. It cannot be the primary one, because a divergence measured on words mostly counts the length of the filing. The divergence is an average surprise less a spread: for any two distributions,

$$\text{KL}(P \parallel Q) = \underbrace{-\sum_k P_k \log Q_k}_{\text{how unexpected these words are}} - \underbrace{(-\sum_k P_k \log P_k)}_{\text{how spread out the filing is}},$$

and the spread of a filing over words is bounded by how many distinct words it has: a description using k terms cannot spread over more than k of them, so its spread cannot exceed $\log k$, while the reference is spread over about 900 terms. A five-word description is charged the whole difference in width, roughly five nats, before any of its content is considered. In 3,000 software-class filings of at least 150 words, scored against one fixed reference and rescored after truncation to a random handful of words, across a fiftyfold cut in length the surprise term moves by 0.016 nats and the spread term by 2.9; synthetic filings drawn at random from the reference itself track $\log(\text{distinct terms})$ to within a fiftieth of a nat (supplementary material). Term-scored atypicality correlates with the log of a filing’s distinct term count at -0.68 in the software class.

Themes break the coupling because the 50 coordinates are always occupied. Under term scoring the effective number of coordinates a filing spreads over rises seventeenfold from the shortest

descriptions to the longest; under themes it varies by less than a third, and not monotonically: a three-word fragment gives the model little evidence, so its inferred proportions stay close to the prior and spread across many themes. The spread term no longer tracks length, and the score is dominated by the surprise term. Theme-scored atypicality correlates with the log term count at -0.22 in the software class and -0.10 in advertising and retail, against -0.68 and -0.62 under terms.

What the themes see, and what they do not. The score compares a filing’s theme proportions to the average proportions of its industry, so it reads which themes a filing draws on and how heavily. It does not, in any material way, read how a filing combines them. Taking each filing’s two leading themes and asking how often that pair occurs together relative to the two themes’ separate frequencies, on a 150,000-filing software-class sample: how rare the themes are individually explains 70% of the variance in atypicality, and the unusualness of the pairing adds under two points on top of it. A filing assembled from ordinary parts in a genuinely new configuration is therefore close to invisible to this measure.

Combinations can be scored in their own right, at two levels, and the two levels run in opposite directions (both computed on the software class, net of the filing’s own lead and atypicality; construction detail in the supplementary material). A *pair lead* — whether the class moved toward the filing’s arrangements of themes — is penalised like lead itself: the top fifth fails the five-year proof of continued use 2.3 points more often than the bottom ($t = 8.4$), and it correlates with lead at only -0.18 . Word-level *recombination* — adjacent word pairs new to the class while both words are established there, novelty assembled from familiar components (Fleming, 2001; Uzzi et al., 2013) — is rewarded: the top fifth fails 2.8 points less with lead and atypicality held ($t = -10.4$), and the two measures are uncorrelated ($r = -0.01$). Arranging themes the industry had not yet arranged is punished like arriving early; coining phrases from the industry’s own words is rewarded. The construct sees neither, and the two run in opposite directions, so its blindness to combination is a bounded omission rather than a hidden bias in one direction.

3.5 Within class, not across it

Goods/services descriptions are drafted to secure scope of protection, not to describe a product distinctively, and counsel reuse language from the USPTO’s own Acceptable Identification of Goods and Services Manual, from house precedent, and from competitors’ successful filings. Identical wording across unrelated firms is therefore common, and it carries through to the scores: 17% of scored registrations issued 2002–2018 share a value with another filing in the same class and registration year. Within a class, the measure picks up departure from a shared drafting convention. Across classes it does not: a class where the Manual supplies dense standard language shows a compressed distribution of atypicality, and one where firms describe novel services in prose a wide one, for reasons unrelated to how innovative either industry is. Every estimate is therefore taken within class and year, and magnitudes compare across classes only as signs and orderings.

The vocabulary is positional rather than causal. A leading filing was ahead of its industry’s language, which implies a trajectory without asserting the filing caused it. Atypicality is textual, and is not trademark law’s distinctiveness doctrine, which governs registrability along the generic-to-fanciful spectrum; the two attach to different objects, one to the mark and one to the description of the goods it covers.

3.6 The gates, the samples, and the conventions

The outcome sections that follow are written to be read independently, so the record’s steps, the outcome names, and the counting conventions are set out once, here.

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a sworn statement of use is filed — the proof that the mark has entered commerce — the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period — the *five-year proof of continued use*, the paper’s central outcome; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a proof is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific proof by its date rather than inferred from a status snapshot.²

Table 3: The trademark funnel in this corpus. Registration counts are unique serials; five-year-proof outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Source: the re-parsed USPTO application backfile, scored as in this section.

Stage	<i>n</i>	Share
Debut filings scored, 1995–2018 (owners’ first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the five-year proof of continued use (event-dated)		52.3%
Passed the five-year proof, cohorts 2002–2013, status known	1,129,724	
failed at the year-ten renewal		39.1%

Three named outcomes recur (Table 3). *Failing to register*: the application never completed, which happens by the applicant’s own inaction in roughly nine cases of ten (Section 4). *Failing the five-year proof of continued use*: a registration cancelled for non-use at registration age 4.0 to 8.5 years, which means the product the mark named left commerce within six years (Sections 5 and 6). *Reaching the SEC record*: the owner appears in dated SEC events — a priced Form D round, financial reporting, a listing (Section 7).

Outcomes are rates, so every estimate is a shift in a probability and is stated in percentage points against the base rate it shifts: +1.4pp on a 52.3% base means 53.7% against 52.3%. The basic comparison is the outcome rate of the fifth of filings with the highest lead minus that of the fifth with the lowest, with the fifths cut inside a single Nice class and year so that no difference between industries or between years enters; regression tables add drafting and filer controls and cluster errors on the owner. Where a continuous version is given, it is the change in the outcome for a one-standard-deviation change in the axis, in percentage points.

²Cross-checked against the USPTO status-code dictionary: 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

Counting differs by table. A filing may claim several Nice classes at once, so a *class-record* counts it once per class while a *registration* counts it once by serial number; a *debut filing* is an owner’s first, one per owner. A *cohort* is the set of registrations issued in a calendar year; where a split is instead by filing year, the text says so.

Two windows bound the samples, each set by an institutional fact. Scoring runs 1995–2019: a full five-year past reference does not exist before 1990 and a full forward reference does not exist after 2020, and the 1995 cut is about two years more conservative than the measured burn-in requires (Appendix A). Five-year-proof cohorts run 2002–2018: cancellations post at about registration age six and a half, the record ends in April 2026, so 2018 is the last cohort whose window has essentially elapsed, and 2002 is the conventional lower cut for this corpus — a bound that matters, because carried back to the first scoreable cohorts the penalty was larger than it has ever been since (Section 6.1).

4 Filing, drafting, and reaching registration

This section is about the first step only: whether an application reaches registration, on every class-record filed 1995–2018. Later steps — whether the product stayed in commerce, and what became of the firm — are Sections 5–7 and do not enter here.

4.1 Registration rises with atypicality and is nearly flat in lead

Figure 1 gives the registration rate by decile of each axis, with deciles cut within Nice class and filing year so that no difference between industries or between years enters. On atypicality the rate rises from 56.0% in the least atypical decile to 59.1% in the ninth and turns down to 58.3% in the most atypical — a three-point climb and a one-point turn at the top. The turn is larger in some industries than the pooled line shows: software and electronics rises through the whole range (51.8% to 60.6%), clothing dips in the middle deciles, and both end higher than they start. On lead the range is 0.9 points with no consistent direction: filings written in language their class was moving toward register at nearly the same rate as filings written in language it was leaving.

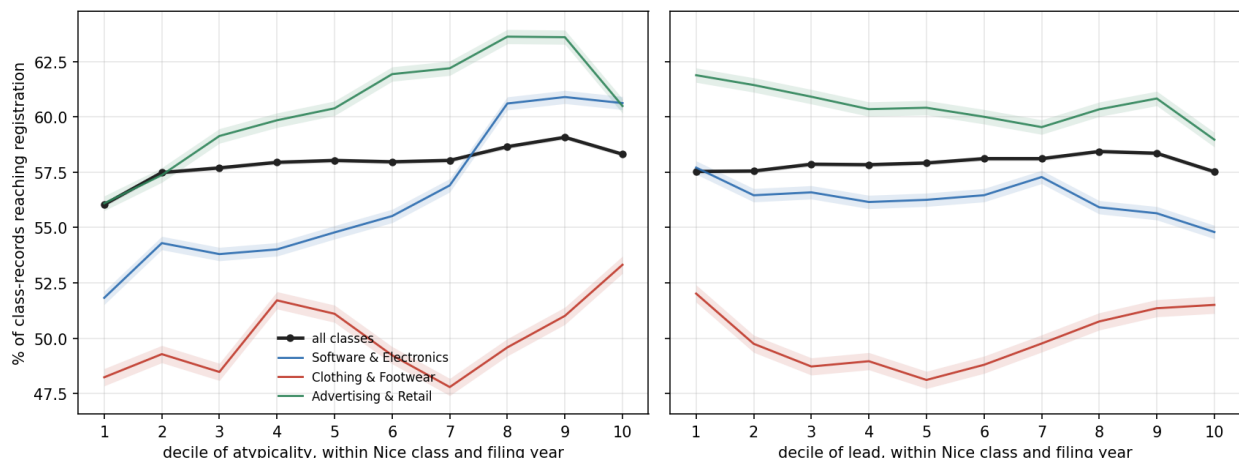


Figure 1: Share of class-records reaching registration, filings 1995–2018, by decile of atypicality (left) and of lead (right), deciles cut within Nice class and filing year; $n = 9,024,532$ scored class-records. The black line pools all 45 classes; the colored lines are three named classes. The grey band is a 95% binomial interval on the pooled line.

The flatness of lead at this step has a mechanical reading worth stating: lead is computed from the five years of filings *after* the filing date, which had not happened when the examiner acted. No participant in prosecution could observe the quantity. Whatever small association appears must run through correlates that were visible at the time — the words themselves, the industry, the drafter. Atypicality’s past-facing half, by contrast, is exactly what an examiner steeped in the class’s usual language experiences, and it is the axis registration responds to.

Figure 2 shows the same outcome on the two surprise axes directly, for the two largest classes: each hexagonal cell is colored by the share of its filings that registered. The color gradient runs along the diagonal — from the low-surprise corner, where registration is lowest, up through the middle of the range — and barely across it. Distance from the diagonal is lead; position along it is atypicality. The plane says graphically what the deciles say numerically: the examination step reads how far a filing sits from its class’s usual language, not which direction it faces in time.

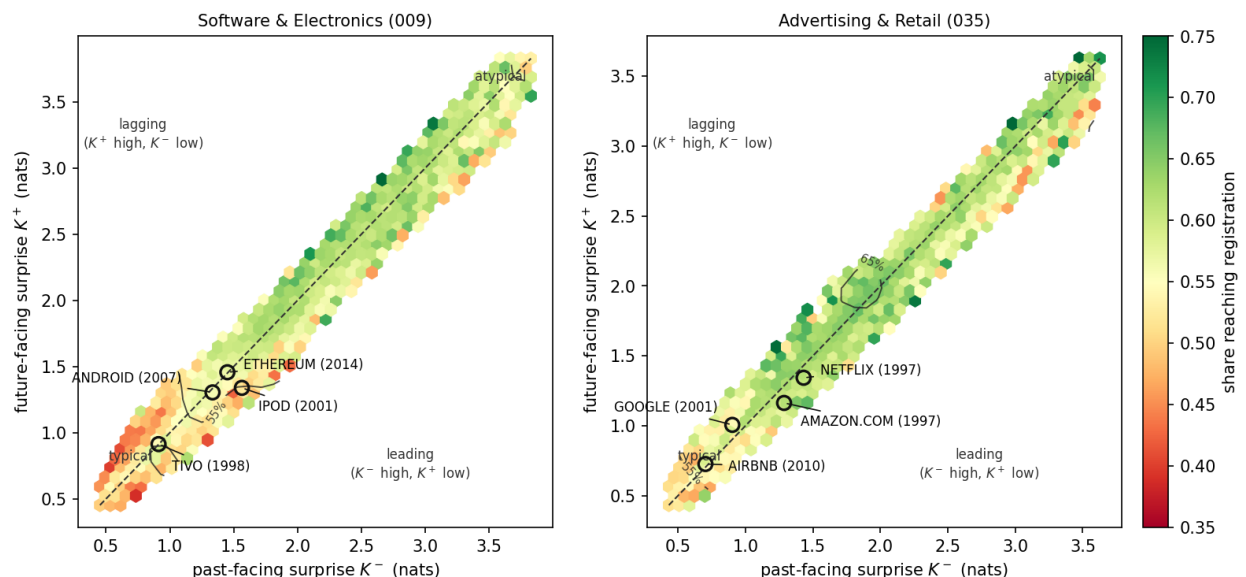


Figure 2: Share reaching registration across the surprise plane, filings 1995–2018 in the two largest classes. Each hexagon holds at least 50 class-records and is colored by the share of them that registered; the dashed line is $K^- = K^+$, on which lead is zero. Cells below the line are leading, cells above lagging.

Two cautions on reading the level. First, registration is mostly a measure of the applicant’s own follow-through: of the applications that do not register, roughly nine in ten die by inaction — the applicant never files the statement of use, or stops responding during prosecution — and only 4.3% are removed adversarially (Table 3). Second, cut on the pooled distribution rather than within class and year, and restricted to debut filings, the atypicality curve is an inverse-U with a deeper conventional tail (46.6% at the bottom, about 58% in the middle, 53.4% at the top); that version carries class and cohort composition, and the supplementary material reports it with the reasons familiarity alone cannot produce the shape. The within-cell profile of Figure 1 is the cleaner statement, and on it the conventional tail, not the atypical one, is the penalized end. The shape is the one audience-conformity research predicts for evaluators (Hsu, 2006; Zhao et al., 2017; Zuckerman, 1999), here from an examiner and at census scale; Tauscher & Rothe (2021) find settings where the optimum sits off-centre.

4.2 Self-filed and professionally filed applications

Counsel of record is observed on every filing. Represented applications are 73% of filings over 1995–2018, falling from 85% in 1995 to 68% in 2018 as online self-filing spread. The two populations file differently and fare differently:

	Share of filings	Reached registration	Median terms (009 / 035)
Represented	73.3%	61.9%	28 / 26
Self-filed	26.7%	45.6%	21 / 17

Self-filers write shorter descriptions (a median of 21 terms against 28 in the software class, 17 against 26 in advertising and retail), sit closer to their class’s usual language, and lean slightly further ahead of it. Because who drafts moves both the language and the outcome at once, counsel of record is held in every regression in the paper, and the later sections report their estimates separately by stratum. The supplementary material draws the full flow from filing to registration split by drafter.

4.3 Office actions and refiling: applicants rewrite toward the class

An application that is abandoned is usually the end of that attempt: among 2.28 million abandoned applications, 3.6% are followed within six years by the same owner filing the same mark again in the same class, and 1.5% by one that then registers. The refilings that do occur show what an applicant changes. Of the 42,115 abandoned applications whose owner later registered the same mark in the same class, 74% carry a different goods/services description the second time. Scoring each registration on the text it was first tried with and on the text it registered with gives the direction of the change. Both self-filers and represented owners rewrite toward the class’s typical description: originals in the most atypical fifth lose 0.39 nats of atypicality, and originals in the most conventional fifth gain 0.16. Refilings that reuse the identical text move by less than 0.01 at every position, so the convergence is in the writing. Who drafts changes the extent but not the direction: owners who hire counsel for the second attempt rewrite most often (82%) and their descriptions lengthen by 16%; counsel rewriting their own earlier draft rewrite least often (72%) and shorten by 4% (supplementary material). Lead barely moves once the reference windows’ shift with the later filing date is taken out.

4.4 What the registration step selects

Three facts carry forward. Registration is a follow-through indicator, not an examiner’s verdict: nine in ten failures are the applicant’s own inaction. Examination rewards a middling specificity, so the rate rises with atypicality, turns down only at the extreme, and is indifferent to lead, which no one at this stage could observe. And applicants who try again conform, from both tails — and an applicant who conformed before ever filing is unobserved, so that part of the compression cannot be bounded. Every estimate on registered marks in the sections that follow is an estimate on this partly conformed language.

5 Persistence in commerce: the five-year proof of continued use

This section is about one outcome: whether a registered mark’s owner proved, between the fifth and sixth year of the registration, that the product the mark names was still in commerce. The sample

is every unique registration issued 2002–2018 that carries a score; failure is a dated cancellation for non-use at registration age 4.0–8.5 years (Section 3.6). Half of registered marks fail it. How the penalty found here moved across industries and decades is Section 6; what became of the firms is Section 7.

5.1 Why the analysis conditions on registration

The question is about companies whose products entered commerce: given that a product was brought to market, does the language of its first description predict whether it was still there five years on? Registration is the record’s marker of entry — a use-based application asserts the mark is in commerce, and an intent-to-use application registers only after a sworn statement of use — so the analysis begins there. An application that never registered left the record, nine times in ten, by the applicant’s own inaction, before any product met a customer. Including those applications would change the question into one about everyone who considered launching, including those who then did not, and since they are 2.3 million of the 6.0 million scored applications, their inclusion also buries the commercial signal in launch noise. Section 9 shows the pooled analysis for a reader skeptical of the conditioning: the registration step’s shape is stamped onto every later outcome, and the gradients found below are attenuated and in places reversed.

Conditioning would miscount if abandonment were usually a pause rather than an ending — if owners reworked refused applications and refiled, the same product would be counted once as a failure and again as a success. It is not: among 2.28 million abandoned applications, 3.6% are refiled within six years and 1.5% lead to a registration (Section 4.3). Crediting every successful refile as an eventual registration moves the registration profile by less than a tenth of a point. The refiled cases are 1.1% of this section’s sample, they fail the five-year proof at nearly the sample rate (51.7% against 52.8%), and the headline contrast below is +2.25pp with them and without them.

5.2 Leading marks fail the five-year proof more often

Marks written in language their industry subsequently moved toward fail the five-year proof more often than marks written in language it was moving away from. Within Nice class and registration year, controlling for whether the filer drafted the mark themselves or hired counsel, and for the length of the description, registrations in the most leading fifth of lead fail 1.4 points more often (53.3%) than those in the most lagging fifth (51.9%; $t = 8.3$, Table 4).

Among 4.12 million registration class-records 2002–2018, dated cancellation at the five-year proof rises from 50.0% at the bottom lead quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 3). But a raw pooled contrast mixes the within-industry gradient with class and registration-year composition, treats multi-class registrations as independent observations, and ignores that outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with lead quintiles cut *within* Nice class and registration year. The specification adds class×registration-year fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;³ and standard errors clustered on 1.27 million

³A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned, which is why each enters as a control.

normalized owners (Table 4).⁴

Table 4: Failure at the five-year proof of continued use and lead: design-based estimates. LPM of event-dated failure on within-class×registration-year lead quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×registration-year fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample.

Sample	Specification	Q5–Q1 (pp)	SE	<i>n</i>
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per sd of lead	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per sd of lead	+0.63	0.06	1,888,352

Standard errors replace significance stars: with *n* in the millions every coefficient has $p < 0.01$, so stars would mark each row identically. Errors are clustered on *normalized owner* — names uppercased, stripped of punctuation and of two dozen legal suffixes, so ACME CORP. and ACME CORPORATION are one cluster. A single owner files many marks, drafts them in a house style, and abandons them together when the business fails; the within-owner correlation of outcomes at the five-year proof is 0.38 pooled across classes (0.42 in the single-class checks where it was first measured), so the 1.27 million owner clusters, not the filings, carry the information. The Madrid row drops §66(a) registrations rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger without them.

The gradient is monotone within cells. Whether the filing was drafted by counsel or by the owner is held because it moves both sides of the comparison at once: self-filers fail the five-year proof at 69.6% against 46.8% for represented owners, and Section 4.2 showed they write shorter, more class-typical, slightly more leading descriptions, so without the control part of any lead gradient would be a self-filing gradient. With it, the penalty is of the same order in both strata (+1.8pp self-filed, +1.2pp counselled). The full control set cuts the magnitude by about a third without absorbing it, and the estimate is if anything *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose failure rates are high for their own reasons.⁵ About forty percent of the raw pooled contrast is composition: +1.4pp design-based against +2.5pp raw. The penalty is also not constant over the sample: within counsel-represented US-domiciled owners the earliest third of cohorts carries none of it (−0.25pp, SE 0.33) — leading marks registered 2002–2007 were no less likely to survive than lagging ones — while the later thirds carry +2.00 and +2.65pp. Section 6 takes that variation as its subject.

⁴Tied scores are common (17.1% of registrations share a lead value within their cell); tie-breaking is deterministic and immaterial, moving the contrast by at most 0.002pp against 2.287 (supplementary material).

⁵The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp.

The per-class extremes of the raw screen are in the supplementary material and the online appendix.

The penalty is a rate, not a timing, and not disengagement. The proof is close to a fixed appointment (96.1% of cancellations post at registration age 6.5–7.0), so a mark failing outside the 4.0–8.5 window would count as a survivor; on the 3.15 million registrations whose ninth anniversary is inside the record, where failure is settled and no window is needed, the contrast is +2.71pp against the window’s +2.65, and conditional on failing, failing late carries a contrast of -0.01pp ($t = -0.2$). Nor is the proof measuring an owner who lost interest: passage is flat across office-action response-speed quintiles (46.1% to 48.1%, medians seven days to 183) and the penalty is present in both speed halves (-3.48 and -2.36pp on passing, against -2.94 pooled). The supplementary material carries the detail, including why 2019 is the last observable cohort.

5.3 The penalty under three theme scorings

Every estimate above scored filings on fifty themes fitted once over the whole corpus, with each class supplying its own reference. That is one of three ways to partition the vocabulary, and the penalty does not depend on the choice. The identical registrations were re-scored under *500 global themes* (the same construction at ten times the resolution, so a theme is closer to a product category than to an industry) and under *50 themes fitted per class* (a separate model on each class’s own filings, so nothing is shared across classes). In all three, surprise is measured against the filing’s own class on per-filing windows of 1,826 days each side.

The lead penalty is a property of the filing, not of the partition (Figure 4, left): +2.29pp under the production scoring, +2.61pp at 500 themes, and +1.94pp under per-class themes, each with a standard error of 0.09, and positive under all three at once in 28 of the 44 classes with at least 3,000 sampled registrations. The scorings agree only moderately filing by filing — lead correlates at 0.64 between the two global resolutions and at 0.52 between the global and the per-class fifty — so a third to a half of the variance in any one scoring’s lead is particular to that scoring, and the penalty survives the disagreement. (Table 5 tabulates the contrasts; the standard errors here are two-sample binomial and unclustered, and outcomes correlate within owner at 0.38, so single-class t -statistics overstate precision by roughly a factor of two.)

Table 5: Failure contrasts at the five-year proof under three theme scorings, on the identical registrations. Top-minus-bottom quintile, quintiles cut within Nice class and registration year, no controls; $n = 3,105,048$, base failure 52.3%.

Scoring	Lead Q5–Q1 (pp)	SE	Atypicality Q5–Q1 (pp)	SE
Global, 50 themes	+2.29	0.09	–4.39	0.09
Global, 500 themes	+2.61	0.09	–4.29	0.09
Per class, 50 themes	+1.94	0.09	–0.72	0.09

Atypicality behaves differently (Figure 4, right). Under the two global scorings the most atypical fifth of registrations fails the five-year proof 4.3–4.4 points less often than the least atypical; under per-class themes the gap is 0.7 points. The two global scorings’ atypicality correlates at 0.79; global and per-class atypicality correlate at 0.39. A global model places every class in one set of coordinates, so a filing that imports a theme the class rarely uses and other classes use often registers as atypical for its class. A model fitted on the class alone re-partitions the class’s own vocabulary, and the imported theme becomes a local theme like any other. What the global scoring calls atypical is, in large part, drawing on a theme from elsewhere.

That population can be measured directly, and it separates two kinds of new. Under the 500-theme model, for each theme and class, the first sustained year is the first year by which the theme has been the dominant theme of at least one hundred filings in that class; the corpus-wide first sustained year is the earliest of those across classes.⁶ Each registration in the sample is assigned its dominant theme and classified at its filing date as carrying a theme new to the corpus, a theme new to its class (established elsewhere, arriving here), or an established theme. Of 4.12 million registration class-records, 3.97 million carry an established theme, 151,012 a theme new to their class, and 779 a theme new to the corpus — the last too few to support a claim, because almost every theme a 500-theme model can resolve had been established in some class before 2002.

Registrations carrying a theme new to their class fail the five-year proof at 48.3% against 51.6% for established themes, and 0.43 points less often within class and registration year (SE 0.13). Holding the filing’s own lead and atypicality as well reverses the sign: net of them, the new-to-class registrations fail 1.06 points more often. The two statements describe one population. A theme arrives in a class in filings that are atypical for that class, and atypical filings survive; that is the -0.43 . Among filings of equal atypicality, the ones carrying the arriving theme fail more; that is the $+1.06$. The atypicality protection of the global scorings is therefore mostly the protection of extension: language established elsewhere and extended into a new class survives. The penalty attaches to being early with language, not to being far from the class’s centre.

Nor is the lead penalty an atypicality effect in disguise. The two could be confounded mechanically — lead is the difference of two nonnegative levels whose average is atypicality, so its spread grows with atypicality and its magnitude correlates with it at $+0.29$ — but the signed value does not (-0.05), and cutting lead quintiles within fifths of atypicality leaves the penalty intact in every fifth: $+2.1$, $+3.1$, $+3.3$, $+1.7$ and $+2.5$ pp from the least atypical fifth to the most (each SE under 0.3).

5.4 The risk persists at the year-ten renewal

Survivors of the five-year proof face a combined declaration and renewal at year ten, and on the design-based estimates the lead penalty does not attenuate there: -0.49 pp per standard deviation on passing the five-year proof, -0.63 pp on renewing at ten (Table 10, both stated on survival). Pooled across classes the renewal profile is a V rather than a slope — renewal is highest just below the centre of the lead axis and lowest at both tails, the lagging tail lowest of all (45.5% against 50.6% at the most leading; fit detail in the supplementary material). Most of the difference between leading and lagging resolves at the five-year proof; what remains at renewal is a penalty on both extremes, the lagging one the larger.

5.5 What the five-year proof shows

Among products that entered commerce, being described in language the industry was moving toward predicts being gone within six years: $+1.4$ points on a 52.3% base with the full design, $+2.3$ raw, in three of every four classes, under every partition of the vocabulary, robust to timing, to drafting, and to who files. Atypical language predicts the opposite, but that protection belongs mostly to filings extending a theme established elsewhere into a new class; net of atypicality, even those arriving-theme filings fail more. The market’s step and the examiner’s step read the text differently for their own reasons: the examiner reads distance from the class’s usual language at a

⁶Sustained years and theme shares are computed on a 25% systematic sample of all filings (every fourth serial), scaled up, over filing years 1986–2024.

moment when lead does not yet exist to be read (Section 4.1); the five-year proof is recorded after the industry’s language has moved, and it is where being early shows up as a cost.

6 New industries, fads, and waves

This section asks when and where the leading penalty of Section 5 lived: across registration cohorts, across the technology classes, across the themes each era’s leading filings were drawn from, in the filings that name the internet, and inside surging themes. The outcome throughout is the same one — failure at the five-year proof of continued use — and every sample is settled cohorts (registrations whose ninth anniversary is inside the record), so nothing here is censored.

6.1 The penalty weakened in the middle of the sample and was largest at the start

Scores exist from 1995, which makes registration cohorts from 1996 evaluable, and their outcomes at the five-year proof have been settled for two decades. Figure 5 draws, for each cohort, the change in survival for a one-standard-deviation increase in lead and in atypicality, the two entered jointly and standardized within Nice class and cohort. Pooled across classes, the lead coefficient is significantly negative in every settled cohort except 2002–2005 — -1.9 to -2.3 pp per standard deviation for the 1996–2000 cohorts, $+0.4$ to -0.2 across 2002–2005, and -0.5 to -1.4 from 2006 on. In the software class the middle of the sample is not a pause but a reversal: the coefficient runs from -6.8 pp for the 1996 cohort to significantly *positive* ($+0.9$ to $+1.2$) in 2002, 2003 and 2005, then back through -4.0 in 2012. The atypicality coefficient never changes sign anywhere: more atypical registrations survive more in every cohort of both panels, by $+0.6$ to $+2.5$ pp per standard deviation pooled and $+2.9$ to $+5.7$ in software.

Stated as cohort contrasts, the swing runs from $+6.9$ pp for the 1996 cohort through $+10.1$ pp for 1999 — roughly twice the largest cohort contrast observed since — to the collapse, and back to $+1.8$ to $+5.6$ pp for every cohort from 2008. Neither prosecution-speed selection in the earliest cohorts nor the cohort indexing drives it: holding pendency in a common band leaves the series unchanged, and indexed by filing year the contrast is $+7.9$ to $+9.6$ pp for 1995–1999 filings against -0.95 for filings made in 2000 (supplementary material).

Splitting the four classes that carry most technology filings — 009, 035, 038 and 042 — against the other forty-one separates a large moving component from a small steady one (Figure 6):

Filing years	Technology classes	All others
1995–1999	+14.18 (0.36)	+4.48 (0.27)
2000–2004	−1.37 (0.35)	+1.21 (0.24)
2005–2007	+2.04 (0.43)	+1.99 (0.28)
2008–2014	+9.02 (0.25)	+1.32 (0.17)

Outside technology the penalty is a flat $+1$ to $+4$ pp in every era, including the one in which the pooled series reads as zero. Inside technology it swings by fifteen points: strongly positive for filings made into the late 1990s, negative for those made in the five years after, and positive again from 2008. The pooled quiet of 2002–2005 is a technology reversal partly offsetting a non-technology positive, not an absence.

What this design cannot say is which cycle the swing belongs to. Marks filed 1995–1999 reach their five-year proof in 2001–2007 and marks filed 2000–2004 reach theirs in 2006–2013, so a story in which unusual language is punished because it was written during an investment boom and a story in which it is punished because the proof falls due during a bust both fit these numbers.

6.2 Three of the four classes reverse, and the leading language changes theme

The four-class split pools industries that do not move together. Taking them one at a time, and then assigning quintiles inside progressively finer cells (Table 6):

Table 6: The leading penalty at the five-year proof inside technology, by class and by cell. Top-minus-bottom lead-quintile contrast in failure (pp, SE in parentheses), settled registrations, by filing era. The upper block assigns quintiles within each class and era. The lower block pools the four classes and assigns quintiles within the era (raw), within class $\times$ registration-cohort cells, and within theme $\times$ class $\times$ registration-year cells, where a filing’s theme is the one its description loads most heavily on under the production $T = 50$ model. $n = 1,248,591$.

Class	1995-1999	2000-2004	2005-2007	2008-2014
009 Electrical, software	+18.04 (0.52)	−5.32 (0.51)	+0.25 (0.62)	+11.67 (0.37)
035 Business services	+4.59 (0.69)	−9.70 (0.56)	+2.02 (0.63)	+1.66 (0.36)
038 Telecommunications	+9.91 (1.40)	−4.24 (1.29)	+2.60 (1.48)	+3.42 (0.94)
042 Scientific, computer services	+17.11 (0.57)	+6.70 (0.62)	+8.42 (0.87)	+12.63 (0.47)
Four classes pooled, raw	+14.49 (0.33)	−1.81 (0.31)	+2.56 (0.38)	+9.16 (0.22)
within class $\times$ cohort	+14.27 (0.33)	−3.63 (0.31)	+2.73 (0.38)	+7.24 (0.22)
within theme $\times$ class $\times$ cohort	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

Electrical and software goods (009) and telecommunications (038) follow the pooled pattern. Business services (035), the class that holds online retail, reverses hardest: −9.7pp for filings made 2000–2004 against +4.6pp in the five years before. Scientific and computer services (042) never reverses. Its contrast is +6.7pp in the era in which the other three are negative and between +8 and +17pp in every other, and by registration cohort it never falls below +3pp (Figure 7). The reversal is an event in goods and in business services; the class that sells technology as a service sat it out.

Assigning quintiles within class and registration year rather than within the era pool changes little. Assigning them within theme, class and registration year changes a great deal: the late-1990s contrast falls from +14.5 to +4.8pp, the post-2008 contrast from +9.2 to +4.5pp, and the 2000–2004 reversal from −3.6 to −1.3pp. Two thirds of the first and half of the second are between themes rather than within them. The leading fifth of each era was drawn from particular themes, and those themes carried their own failure rates; the penalty for being leading *inside* a theme is a steadier +2.5 to +5pp, the same order as the penalty outside technology.

Which themes is a matter of record (Table 7; the table’s last row is the all-themes pooled contrast within theme $\times$ class $\times$ cohort). In 1995–1999 the computer, software, information and data services theme held 38% of technology filings and 75% of the leading fifth, and failed at 64.9% — that one theme is most of the late-1990s penalty. In 2000–2004 the leading language had moved to electronic media while the internet-services wording became lagging, and inside several themes the lagging fifth now failed more than the leading one. From 2008 the leading fifth over-represents downloadable and online software four to one — the application store’s vocabulary, the highest-failure theme (67.2%) with the largest within-theme contrast (+8.4pp pooled across cohorts). The full theme tour is in the supplementary material.

So the swing has two layers. Across the whole period, a leading filing is a few points more likely to fail than a lagging one in the same theme, class and cohort, in technology as elsewhere. On top of that, each era’s leading vocabulary was concentrated in one or two themes whose marks failed at their own rate: internet services in the 1990s, application software after 2008. In the trough

Table 7: The leading penalty at the five-year proof by theme within the four technology classes. Themes with at least 20,000 settled registrations, labelled by their five highest-probability terms; quintiles assigned within theme and era. Cells with fewer than 3,000 registrations are blank. The final row pools all themes, with quintiles cut within theme×class×cohort.

Theme (top words)	<i>n</i>	1995-1999	2000-2004	2005-2007	2008-2014
services, computer, software, information, data	451,253	+3.61 (0.52)	+3.56 (0.49)	+4.18 (0.64)	+5.92 (0.37)
video, computer, electronic, music, games	162,290	+2.53 (0.97)	−3.62 (0.89)	−2.85 (1.01)	+3.36 (0.60)
services, featuring, store, retail, store services	131,107	+8.39 (1.09)	−5.49 (1.01)	+2.92 (1.15)	+0.98 (0.66)
services, field, training, estate, educational	95,602	+7.82 (1.09)	+0.99 (1.12)	+6.02 (1.43)	+2.05 (0.83)
apparatus, systems, devices, electronic, control	84,121	+4.80 (1.27)	−6.92 (1.19)	−0.58 (1.41)	+3.65 (0.88)
design, field, systems, development, services	32,517	+10.40 (2.03)	+4.67 (1.82)	+1.50 (2.27)	−2.27 (1.45)
software, online, downloadable, non-downloadable, virtual	29,472	—	—	—	+3.30 (1.05)
bags, clothing, shirts, jackets, leather	28,421	+1.28 (2.59)	−1.55 (2.39)	−4.66 (2.40)	+6.06 (1.36)
research, medical, scientific, testing, purposes	25,897	+12.65 (2.66)	+12.09 (2.03)	−3.82 (2.59)	+5.63 (1.51)
services, health, field, care, information	22,880	+4.94 (2.04)	+1.39 (2.36)	—	+3.50 (1.72)
All themes, within theme×class×cohort	1,150,921	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

between, the language that had been leading became lagging, and the marks still written in it failed with it. The design-based estimates of Table 4 hold class and cohort fixed but not theme, so they carry both layers; a reader who wants only the within-theme penalty should take the lower block of Table 6 as the guide to its size.

6.3 Internet as a theme: web-based filings against their class

A filing is internet-bearing when its goods/services text names the internet, the web, a website, online or on-line provision, or electronic commerce. The rule reads the text rather than the class, so a web retailer filing in apparel is caught and a semiconductor maker filing in class 009 is not. Internet-bearing filings are 16.9% of scored registrations: 58% in telecommunications, 39–41% in advertising and retail, scientific and technical services, and legal and personal services, under 5% in most goods classes. Figure 8 puts every class on two axes at once — how much more often its internet-bearing filings register, and how much more often they fail the five-year proof; the supplementary material tabulates the underlying rates for every class.

Three patterns are in the figure. In the industrial goods classes, internet-bearing registrations fail the five-year proof far more often than their class: machinery 50.9% against 38.7%, chemicals 46.1% against 39.1%, vehicles 49.6% against 45.1%. A web-based business registering in an engineering class fails at the rate of a software registration (51.4%), not at the rate of its class. In the consumer goods classes the sign reverses — internet-bearing registrations in clothing fail 52.4% against 58.7% for the class, in household utensils 44.5% against 50.5% — so online sellers of consumer goods outlast the class’s other registrants. In every service class the internet-bearing filings fail more often, by one to eleven points, most in legal and personal services (60.1% against 48.7%).

At the registration step the industrial-goods classes sit to the right of zero: internet-bearing applications there reach registration 59.5% of the time against 56.0% for the rest. For those classes the two steps move in opposite directions — more likely to be granted, then more likely to be cancelled for non-use six years later — which is the pattern Section 5.5 predicts when the examiner reads distance from class language and the market reads whether the product stayed in commerce.

The failure gap has a common shape in event time. Dating each class’s internet arrival at the first filing year the pattern reaches 2% of its filings, the gap starts near +7 points in the first observable cohorts and settles between +1 and +4 nine to eleven years after arrival, shrinking in six of the eight classes observable in both phases; calendar time disguises this, since machinery’s arrival was 2008 and its peak 2012–2015 (supplementary material). The class’s clock starts when the vocabulary arrives, not when the internet did — the shape falling entry costs followed by normalization would leave.

The internet split also locates the era swing of Section 6.1. Rerunning the per-cohort lead coefficients of Figure 5 separately for internet-bearing and other registrations (Figure 9): the mid-sample reversal belongs to the internet-bearing filings, whose leading registrations survived *more* in the 2001–2005 cohorts (+1.3 to +2.5pp per standard deviation) while non-internet technology registrations sat near zero. The other two episodes are not the internet’s in this textual sense. The late-1990s penalty is present in technology registrations that never name the web (−3.4 to −4.8pp for the 1996–2000 cohorts) — consistent with Table 7, where that era’s deadly vocabulary is computer and software services, words that predate explicit internet naming. And the post-2008 penalty sits in both splits at once (−2 to −3pp each). Outside technology, the non-internet series is the steady −0.4 to −1.4pp throughout. What the internet changed, on this cut, is the middle: for half a decade after the crash, the web filings whose language led their class were the ones that lasted.

The Nice Classification has twice faced this question and twice declined to make the internet a class.⁷ The survival pattern bears on the choice: internet-bearing filings in a goods class fail at rates closer to the technology classes than to their own, so the technology overlay carries more information for this outcome than the class assignment — and Nice classes used as industry controls, as this paper and the literature use them, do not partition the internet economy.

6.4 Surges: joining a wave

Genuinely new themes are rare (Section 5.3); the observable event is a surge, an established theme’s share doubling, over its own three prior years, to a level that matters. The one corpus-scale episode — online software doubling to 2.4% of all filings in 1999 — failed the five-year proof 8.6 points above class and year, 8.5 net of the filings’ own lead and atypicality, with no gradient in arrival order inside the wave (−0.2pp, SE 1.9). Class-scale surges are common (962 episodes at a 1% within-class threshold) and cost little pooled: +0.65pp within class and year, +1.27 net of lead and atypicality, with episodes differing in sign among themselves. Within waves, entry order does not matter; larger waves are not deadlier; and growing waves carry more excess failure than receding ones (+4.3 against +1.0 points, suggestive on 61 episodes) — a pattern consistent with selection at the proof rather than crowding out. The surge figure, the episode table, and the wave-level facts are in the supplementary material.

⁷The eighth edition (2002) split the old class 42 into classes 42–45 by kind of service; the twelfth (2023) placed downloadable virtual goods in class 9 and distributed metaverse services across the existing service classes.

6.5 What the waves show

The leading penalty is steady and small outside technology and episodic inside it, and most of the episode is between themes: each era’s leading vocabulary was concentrated in one or two themes that failed at their own rate, while the within-theme penalty holds at +2.5 to +5pp throughout. Web-based filings fail hardest in each class’s own first internet years and converge toward their class within a decade. Joining a surging theme costs little on average and nothing for arriving early within the wave; the one corpus-scale wave, online software in 1999, is the exception at every margin. What a fad costs, on this record, is mostly a matter of which theme the fad was, not of when the filer joined it.

7 Commercial success: funding, listing, and the firms that mattered

A mark that survived the five-year proof names a product that held its place in commerce. This section asks whether the owners of leading or atypical debut filings went on to the outcomes that make a firm rich: a priced private round, SEC financial reporting, a public listing, and revenue at scale. The unit is the owner’s first registered filing, its debut, scored on the production measure; the outcomes are dated events in the SEC record. Form D became mandatory in electronic form on 16 March 2009, so the funding ladder is built on debuts made 2009–2018, for which every rung is observed; the public-reporting analysis uses the longer 1995–2018 sample.

7.1 The ladder: a round, reporting, a listing

Of 999,442 owners whose first registered filing, carrying a score, was made 2009–2018, 17,115 (1.71%) raised a priced private round after the debut, 4,383 (0.44%) appear in SEC reporting records at any time, and 1,795 (0.18%) registered a class of securities or completed an initial public offering at any time. The funded rung requires the round to postdate the debut; the reporting and listing rungs do not, so owners already reporting before their first trademark filing are counted, and the staged analysis in Section 8 shows they are a majority of the exchange-listed. The rungs nest: 7.3% of funded owners reach reporting against 0.3% of the unfunded, and 41% of reporting owners reach an IPO.

Table 8 gives the rate at each rung by fifth of lead and of atypicality, with fifths cut within Nice class, debut year, and fifth of description length, so that neither the industry, the year, nor the length of the description enters the contrast. Funding does not vary with either axis: the most leading fifth of debuts is funded at 1.80% against 1.86% for the most lagging, and the most atypical at 1.64% against 1.68%, both inside sampling error. Trademark counts and breadth carry a funding signal (Block et al., 2014); vocabulary position does not. Reporting and listing do vary. Owners whose debut language was most lagging reach reporting at 0.59%, owners whose debut language was most leading at 0.38%; for an IPO the rates are 0.24% and 0.17%. On atypicality the sign is positive on this sample: the most atypical fifth reports at 0.53% against 0.42% for the most conventional, and lists at 0.22% against 0.15%. Section 7.3 absorbs description length continuously on 1995–2018 debuts and finds the atypicality gradient flat; with length held in fifths on 2009–2018 debuts it is positive at t above 4. The two are stated side by side; the difference between them is a length control and a sample, and the atypicality result at the listing margin is the one claim in the paper that depends on which is used.

The lead profile at reporting and at IPO is not monotone: it falls from the lagging end through the fourth fifth and turns up in the fifth. Owners whose first description used language the class

Table 8: The ladder by fifth of debut lead and of debut atypicality. Rates in percent of owners whose first registered filing was made 2009–2018 ($n = 999,442$); fifths cut within Nice class, debut year, and fifth of description length. The last column is the most-minus-least contrast with its t -statistic.

Rung	Axis	Q1	Q2	Q3	Q4	Q5	Q5–Q1 (pp; t)
Funded	lead	1.86	1.73	1.61	1.57	1.80	–0.06 (–1.4)
	atypicality	1.68	1.70	1.79	1.76	1.64	–0.04 (–1.1)
Reporting	lead	0.59	0.49	0.40	0.34	0.38	–0.21 (–9.4)
	atypicality	0.42	0.36	0.43	0.46	0.53	+0.11 (+5.2)
IPO	lead	0.24	0.21	0.14	0.13	0.17	–0.07 (–4.5)
	atypicality	0.15	0.15	0.18	0.21	0.22	+0.07 (+4.8)

was leaving behind are the likeliest to report and to list. Most of them are established firms making a first trademark filing late in a business already built on the class’s older vocabulary, and the ladder cannot separate that from anything the language itself does. Listings that arrive without an observed post-debut Form D round lean the same way: their owners’ debut lead averages -0.25 standard deviations against -0.08 for funded listings — consistent with established, revenue-financed businesses listing without venture money, where “unfunded” means only that no Form D notice is observed.

7.2 The firms that produced the most value

Among the 1.80 million owners whose first registered filing was made 1995–2018, 4,710 link to SEC financial statements with reported revenue; ten of them hold 18.5% of the \$19.2 trillion of linked peak revenue and five hundred hold 82.9%. Reading this handful is not a statistical argument, but it is not a stray anecdote either: these are the firms in which most of the linked value was generated, and their debut language was ordinary — the ten largest sit 0.35 standard deviations *below* average atypicality and within 0.01 of average lead, and the revenue-weighted linked set sits at the average on both axes (tier detail in the supplementary material). Hand-curated vocabularies sharpen the same point from the other side: debuts in AI or blockchain vocabulary are funded four to five times as often as the average debut (7.8% and 8.4% against 1.7%), but reach reporting and listing only about twice as often, on small counts (supplementary material).

7.3 Leading debuts reach public reporting less often; unusual ones no more often

The outcome here is whether an owner ever appears in the Securities and Exchange Commission’s financial reporting records. That is broader than exchange-listed: roughly half the matched firms carry a ticker and the rest report for other reasons. Figure 10 draws the raw rate against each axis in twenty bins. In signed lead the profile is a U whose tails both sit above the base rate and whose lagging tail is the higher; in lead magnitude and in atypicality the raw rate rises.

The raw atypicality gradient does not survive the design-based estimates (Table 9). Within class and filing year, holding description length fixed, top-quintile atypicality raises the probability of SEC reporting by +0.000pp on a 0.481% base ($t = 0.0$), and by -0.003 pp on the past level alone. The quintile profile is not even monotone: the three middle quintiles sit *below* the bottom one and the top merely returns to it. The 2.7-fold gradient in the raw bins is composition: firms that go on to report cluster in particular classes, years, and multi-class filings, and class $\times$ year fixed effects

absorb it. Lead costs, at equal atypicality: the signed component enters negatively (-0.047 pp per standard deviation, $t = -7.8$) while the unsigned component enters positively ($+0.067$ pp per standard deviation, $t = 7.7$). The U in the raw bins is exactly this pair: lagging filings show the magnitude premium with no lead penalty, leading filings show it net of that penalty, and the middle bins show neither. Once the levels are controlled, the recovery of the far-leading tail is much reduced: the quintile profile falls through Q4 and recovers only slightly at Q5.

Table 9: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.481%.

	Coefficient (pp)	SE
<i>Atypicality quintiles, $A = \frac{1}{2}(K^- + K^+)$ (Q1 omitted)</i>		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^- alone	−0.003	0.018
<i>Signed lead quintiles, FE + log length + KL levels</i>		
Q3	−0.133	0.020
Q4	−0.187	0.021
Q5	−0.158	0.024
<i>Decomposition, FE + log length</i>		
lead magnitude $ L $ (pp per sd)	+0.067	0.009
signed lead L (pp per sd)	−0.047	0.006
log description length	+0.056	0.004

Standard errors in place of significance stars; see the note to Table 4.

Owners are matched to SEC registrants by normalized exact name, over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants, and a base rate of 0.481% among registered debut filers. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22. A random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate.

The linkage is nonetheless a name match, and that bounds the atypicality reading in particular. Firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text; nothing in this design separates the two. The lead coefficient is unchanged under alternative linkages; the atypicality coefficient is not, so only the lead result is robust to the name-match design. The financing rows of Table 10 rest on a Form D match whose raw extracts are not in the current tree and should be read as provisional.

The U is also not atypicality working through lead’s construction (the mechanical coupling stated in Section 5.3): on this sample lead magnitude and atypicality correlate at $+0.25$, signed lead at only -0.10 , and cutting lead fifths within fifths of atypicality keeps the structure: the most lagging fifth out-reports the most leading in four of the five atypicality strata (by 0.08 to 0.39pp on sub-1% bases), and the ordering reverses only in the most atypical fifth, where the most leading debuts report most. The lagging tail’s premium is not a ceiling artifact; the leading tail’s partial

recovery in the raw U is where the atypicality composition lives, which is what the level controls in Table 9 absorb.

“SEC reporting” is not “went public”: splitting matched owners on whether their SEC filer number carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same U in lead (Section 9), so the result is not an artifact of counting non-operating registrants as successes.

7.4 The construct is not patenting

Within the firms that do both, trademark vocabulary and patenting move independently. On a panel of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024), each axis is regressed on $\log(1 + \text{patents})$ with firms absorbed and errors clustered on the firm, estimated on 94,181 firm-years across 17,506 firms under theme scoring (93,551 and 17,415 under term scoring) — 134,075 matched firm-years before dropping the firm-years that contribute no within-firm variation. The outcome is standardized, so β is the shift in standard deviations of the axis per unit of log-patents; one unit is roughly a tripling of the patent count, and a firm going from none to twenty moves about three units.

Text counted as	Axis	β (sd)	t	Between-firm r
Themes	atypicality	+0.012	+3.0	—
Themes	lead	−0.058	−9.4	−0.020
Terms	atypicality	−0.055	−11.5	—
Terms	lead	+0.012	+2.0	—

The grid is close to a mirror: whichever axis carries the small signal under one way of counting the text carries the opposite sign under the other. The theme-scored *lead* coefficient (−0.058) and the term-scored *atypicality* coefficient (−0.055) are nearly the same number attached to different quantities. Only those two cells are established; the mirror’s other diagonal is within noise, and the orthogonality conclusion rests on the magnitude bound — every cell under 0.06 standard deviations — rather than on any cell’s sign. Nothing here reaches a tenth of a standard deviation.

One objection is timing: a patent is applied for before its invention is in public use and a trademark when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly (Seip et al., 2018), and no timing offset from $t - 2$ to $t + 1$ isolates a direction (−0.054 to −0.089 sd under theme scoring, +0.012 to +0.024 under terms); among owners who both patent and raise a priced round, neither event systematically precedes the other (median lag zero years).

The second objection is heterogeneity: a pooled near-zero could hide a positive slope where the two rights are complements offsetting a negative one where trademarks work alone. On a coarse, judgement-based sorting of classes into complements and substitutes — assigned in advance from the class definitions, nothing estimated — the slopes are −0.000 among complements, −0.026 among substitutes, interaction −0.008 ($t = -0.49$), and the complement classes do not agree in sign with one another (cell detail in the supplementary material). Neither retention supports the complementarity story, with the caveats that a coarse sorting can miss a finer-grained one and that conditioning on a successful assignee match pushes any interaction toward zero.

7.5 Leading does not pay, even where novelty does

The answer to this section’s question is no. An owner whose surviving product had been described ahead of its industry’s language is not more likely to raise a priced round, reaches SEC reporting and listing *less* often than a lagging owner, and is absent from the top of the linked revenue

distribution, whose firms debuted in ordinary language. What pays at the reporting margin is unsigned: lead magnitude and description length, the marks of a specific, seriously drafted first filing, carry positive coefficients while the signed component stays negative. Being unusual is at worst free and at best mildly rewarded; having been early is never rewarded at any rung the record shows, and the listings that arrive without observed venture money belong disproportionately to the lagging.

8 Discussion: each axis bites at a different stage

Table 10 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner’s first filing — under a single specification, so the coefficients can be read down a column. Magnitudes differ from the section-specific tables above.

Table 10: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner’s first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference. Cohort windows differ because each maintenance gate needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The first gate is a dated cancellation for non-use; the second is whether a registration that cleared the first was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten death share a terminal status and are separable only this way. This table alone is estimated at $T = 200$ topics rather than the $T = 50$ used elsewhere; the sign and ordering of every row are unchanged at $T = 50$.

Outcome (<i>population</i>)	<i>n</i>	Base	Unsigned		Signed
			Atyp.	ΔKL	Lead
Reached registration (<i>filed</i>)	3,161,552	56.11%	−1.352 (0.033)	−0.307 (0.044)	+0.872 (0.028)
Passed gate 1, yr 6 (<i>registered</i>)	1,260,392	41.82%	+1.349 (0.054)	−1.150 (0.082)	−0.487 (0.052)
Passed gate 2, yr 10 (<i>gate 1</i>)	297,944	55.92%	+1.296 (0.108)	−0.297 (0.162)	−0.629 (0.105)
Raised Reg D round (<i>filed</i>)	1,535,004	2.53%	+0.003 (0.015)	+0.148 (0.027)	−0.149 (0.016)
Raised Reg D round (<i>registered</i>)	916,448	2.61%	−0.017 (0.020)	+0.115 (0.035)	−0.114 (0.021)
Listed after round (<i>funded</i>)	38,886	3.49%	+0.493 (0.129)	+0.608 (0.170)	−0.513 (0.104)
Became SEC-reporting (<i>registered</i>)	1,773,938	0.56%	+0.042 (0.007)	+0.040 (0.011)	−0.004 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 4.

Surprising language costs at registration and pays afterwards: atypical debut filings complete registration less often (−1.35pp per standard deviation), then survive the five-year proof of continued use more often (+1.35pp) and reach public markets more often (+0.04pp on a 0.56% base — the one specification in which a positive atypicality coefficient on listing survives, and it is a continuous term through a non-monotone profile). Having been early runs the other way: leading applications complete registration *more* often (+0.87pp) and then pass the five-year proof *less* often (−0.49pp on passing). Both survival rows point the same way at the year-ten renewal, four years later and on a sixth of the sample: atypicality +1.30pp and lead −0.63pp among marks that had already cleared the five-year proof. Whatever the five-year proof selects on, the renewal selects on it again rather than reversing it.

In the financing rows atypicality does nothing at one stage and a great deal at the next. Whether a debut owner ever raises a Regulation D round is unrelated to how atypical its filing was (−0.017pp per standard deviation against a 2.61% base, $t = -0.8$); among owners who did raise one, the

same measure predicts *reaching* the public market (+0.49pp on a 3.49% base, $t = 3.8$). Atypical language is invisible at the financing stage and informative afterwards, which is what an investor screen that does not read the text would look like — the text being free and public at the moment of filing. The comparison is among survivors, so it cannot separate that reading from selection on unobservables. Three limits bound these rows: Regulation D notices are electronic only from 2009; owners are matched by normalized name, which succeeds for a minority favouring formally constituted entities; and the final row conditions on funding realized after the filing being scored, so it establishes that information distinguishing eventual listers was present in the text years earlier, not that the language caused the outcome.

Lead does not reverse anywhere after registration: negative at both use proofs, at financing with or without registration imposed, and at listing among the funded; the only row where it is not negative is SEC reporting among registered owners, where it is indistinguishable from zero. Leading filings fare worse at every stage where the market, rather than the examiner, does the selecting.

Why would having been early cost? The two axes are not independent in the obvious economic sense. What makes a filing leading is precisely that its industry moved toward its language, and an industry that has moved toward your language now describes its goods the way you describe yours. Being early therefore consumes the quantity that pays: the mark that was unusual in 2011 is ordinary by 2016 not because its owner changed anything but because the category arrived. On this reading atypicality is a position competitors erode, and lead measures how fast. The design cannot demonstrate that mechanism; what can be said is that a negative coefficient on lead at equal atypicality is what the mechanism would produce if it were real.

8.1 Implications for trademark-based measurement

Three practices follow for research that reads trademark text. First, score distributions, not words. Word-level divergence is dominated by description length — on this corpus the correlation with log distinct terms reaches -0.68 — and firm-level performance correlations built on it dissolve or reverse under distribution scoring on identical samples. Any term-level novelty measure should be accompanied by the theme-level rescoring that costs one model fit. Second, treat the registered description as a conformed document. Applicants who refile rewrite toward their class’s typical language from both tails, whoever drafts the second attempt, so samples of registered marks understate the atypicality applicants first attempted, and estimates on registered text are estimates on prosecution-filtered language. Studies using description similarity or breadth inherit this compression. Third, the two components of surprise are not interchangeable and should not be pooled. Lead is robust to every partition, weighting, and window tried here; atypicality’s apparent effects can be artifacts of the partition (its protective effect at the five-year proof belongs to theme importation, visible only to global models) or of a control (its listing effect depends on how description length is held). A study that reports “novelty” from trademark text without separating the temporal from the cross-sectional component may attribute to one what belongs to the other.

For users of the classification itself, one caution: internet-bearing filings in goods classes survive at rates closer to the technology classes than to their own, so Nice classes used as industry controls do not partition the internet economy. And for the maintenance data, one asset: the five-year proof of continued use is a dated, sworn, product-level survival record covering every registered mark — an outcome the indicator literature has largely left unused, observable for any cohort more than seven years old at no cost beyond parsing the prosecution history.

9 Robustness: what changes under other choices

Each result above is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. The table summarizes what each choice costs; the paragraphs carry the four checks a reader will probe, and the supplementary material carries the full treatment of everything else — functional forms, the term-scored replication and its firm-level hazard, the listing-versus-reporting split, the window-mix decomposition, and the anatomy of L as a small residual of two large levels. No sign anywhere depends on any of these choices.

Variant	Baseline $\rightarrow$ variant	Reading
Terms for themes (flat / decayed)	lead $+2.29 \rightarrow +3.56$ / $+2.92$ pp	conservative representation
Decayed reference windows	$+2.29 \rightarrow +1.95$ pp	the one estimate that moves
Resolution $T \in \{50, 100, 200, 500\}$, class 009	$+3.2 \rightarrow +2.7$ to $+5.7$ pp	sign fixed; magnitude a band
Corpus-wide at $T = 200$	$+2.29 \rightarrow +2.23$ pp	pooling steadies it
Same T , second seed	$+5.20 \rightarrow +4.88$ pp	sweep is mostly fitting noise
Window width 2–7 years	$+2.02$ to $+3.37$ pp	flat from five years
Window mix, foresight / rupture	$+1.65$ / $+3.00$ pp	rupture the stronger sense
Proof window 3.5–9.0 / 4.5–8.0	$+2.29 \rightarrow +2.29$ / $+2.28$ pp	window immaterial
Cohorts 2002–09 / 2010–18	$+2.29 \rightarrow +0.98$ / $+3.11$ pp	the era swing of \$6.1
Excluding duplicate-text filings	$+2.29 \rightarrow +3.05$; atyp. $-4.39 \rightarrow -5.68$ pp	boilerplate dilutes
Excluding Manual-scale texts only	$+2.64$; -5.05 pp	same
Atypicality, cohort halves	$-4.39 \rightarrow -4.55$ / -4.28 pp	stable
Unconditional (no grant filter)	lead composite $22.4 \rightarrow 22.1\%$	conditioning exposes, not creat
Ladder, counsel / self debuts	reporting lead $-0.21 \rightarrow -0.28$ / -0.04 pp	gradient is counsel-borne
Ladder, 2009–13 / 2014–18	reporting atyp. $+0.11 \rightarrow -0.01$ / $+0.20$ pp	atypicality belongs to the late l

Representation and weighting. Four per-filing builds exist: themes or terms, flat or two-year-half-life weighting. Weighting is close to irrelevant (lead correlates at 0.97–0.98 across it, atypicality at 1.000); representation is the real test (themes and terms agree at only 0.46 on lead), and it is the conservative one — term scoring returns larger lead contrasts ($+3.56$ and $+2.92$ pp) and smaller atypicality contrasts (-3.63 , -3.73) than the production build’s $+2.29$ and -4.39 .

Resolution and fitting noise. Re-fitting the theme model at $T \in \{50, 100, 200, 500\}$ leaves the sign and significance of the software-class contrast untouched while the magnitude moves non-monotonically between $+2.7$ and $+5.7$ pp; corpus-wide, the only two resolutions that exist agree at $+2.29$ and $+2.23$ pp. A second fit at $T = 200$ differing only in its random seed agrees with the first at $r = 0.79$ on lead — most of the apparent resolution sensitivity is fitting noise, the estimate moves only $+5.20 \rightarrow +4.88$ pp, and the implied measurement error attenuates every lead coefficient toward zero rather than manufacturing it.

Decayed windows. Weighting recent language up with a two-year half-life is the one choice that shifts the headline: $+1.95$ pp against $+2.29$, the same sign, precision, and profile. The flat window is kept as the simpler statement; a reader who prefers recency should carry $+1.95$.

Conditioning and composition. Without the grant filter, the composite outcome (registered and passed the proof) is nearly flat in lead because two opposed gradients cancel — registration rising, survival falling — so conditioning exposes the market’s gradient rather than creating it. Excluding the 47% of registrations whose exact normalized text is shared with another filing *strengthens* both headline gradients (lead $+3.05$ pp, atypicality -5.68); identical texts carry identical scores and cluster at the typical end, diluting both. The non-monotone profiles on signed lead decompose into a symmetric part inherited from atypicality composition, a symmetric conditional lead-magnitude

effect, and a small linear tilt whose fitted vertices land where that arithmetic puts them (supplementary material).

Two cautions carry across every raw contrast: outcomes correlate within owner at 0.38, so unclustered t -statistics in the per-class, per-era, and per-episode cells overstate precision by roughly a factor of two; and the penalty is not constant over the period, so the pooled row is an average over the swing of Section 6.1, and a reader who wants the penalty for a period should read Figure 5 rather than the pooled estimate.

A Data and scoring parameters

Table 11: Corpus, scoring, and samples in one place. Full construction detail, the per-class breakouts, and all demoted analyses are in the supplementary material and the online appendix.

Corpus	USPTO Trademark Annual Applications backfile (TR-TYRAP): 13.99 million case files, 1884–2025, re-parsed to 242 million dated prosecution events.
Theme model	LDA, $T = 50$, fitted once on a stratified sample of 448,437 filings across all 45 Nice classes; vocabulary of 62,168 unigrams and bigrams; resolutions 500 (global) and 50 (per class) refit for Section 5.3.
References	Per-filing windows of 1,826 days on each side of the filing date, same-class filings only, flat weighting; a filing with fewer than 500 same-class filings on either side is unscored. Scoring years 1995–2019.
Registration sample	9.02 million scored class-records filed 1995–2018.
Five-year-proof sample	3.10 million unique scored registrations issued 2002–2018; failure is a dated §8/§71 cancellation at registration age 4.0–8.5 years.
Firm samples	2.07 million registered debut filings 1995–2018 (SEC reporting); 999,442 debut owners 2009–2018 (funding ladder).
Owner links	SEC registrants and Form D issuers by normalized exact name; PatentsView assignees likewise.

The burn-in cut at 1995 is set empirically: thin past references inflate past-facing surprise, and the measured contamination falls below any interpreted effect size from 1993, so the 1995 cut is conservative by about two years (Figure 11).

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Declaration of competing interest

The author declares no competing interests. The author’s current employment is unrelated to this work, and the views expressed are the author’s alone.

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Data availability

The corpus build, all analysis code, every generated artifact, and an interactive per-class online appendix are public at <https://github.com/ericsilver/tm-vocabulary>. The underlying USPTO, SEC, and PatentsView sources are public.

Declaration of generative AI in the writing process

This paper was produced in a sustained collaboration with Claude (Anthropic), which drafted and revised text and analysis code under the author’s direction. Every analysis decision, every number against its artifact, and all final text were reviewed by the author, who takes full responsibility for the content.

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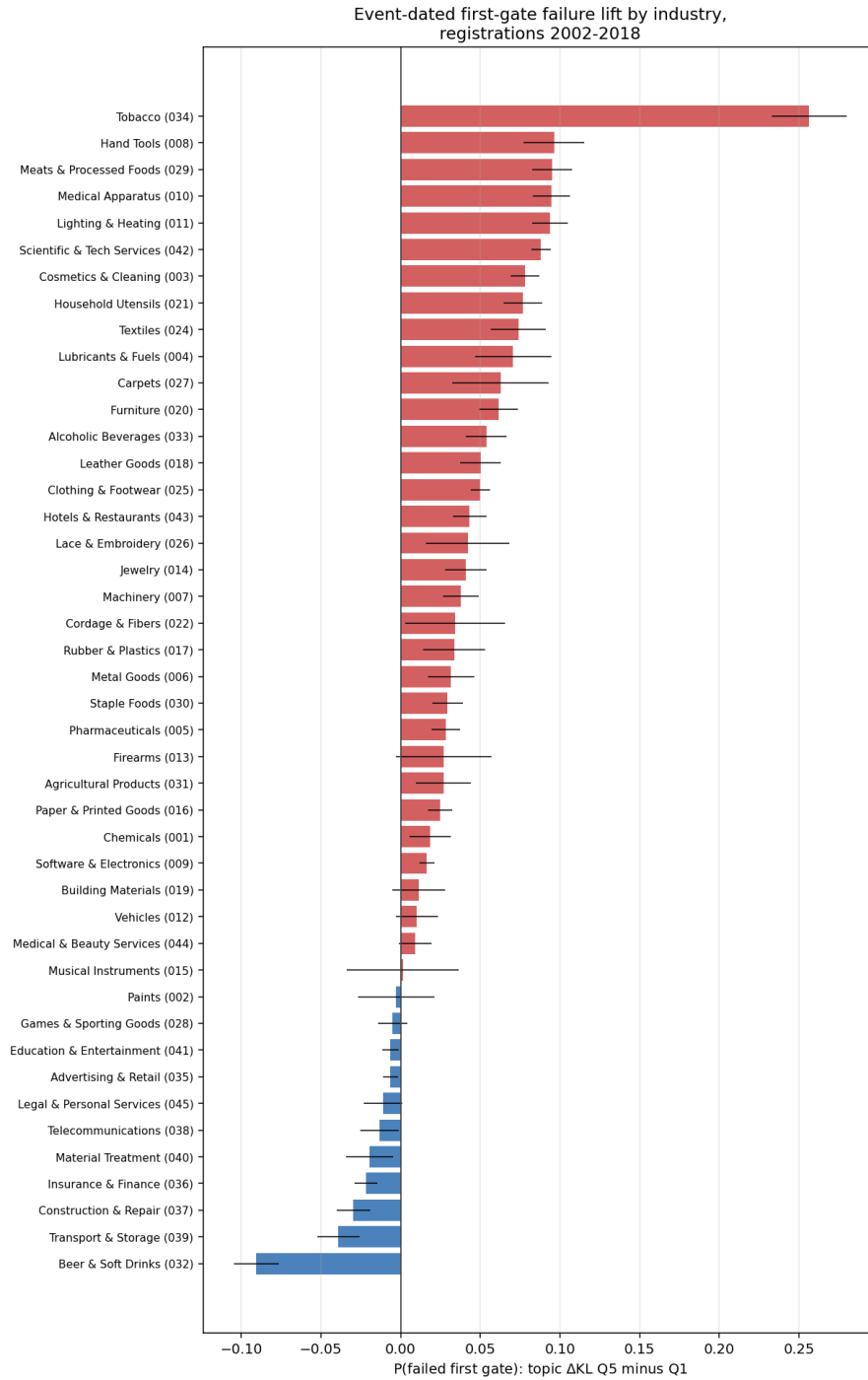


Figure 3: Failure lift at the five-year proof of continued use (lead top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts; classes need 5,000 scored registrations to enter. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals, uncorrected for the within-owner correlation (0.38) and so narrower than clustered intervals; read Table 4 for precision. Per-class breakouts are in the online appendix.

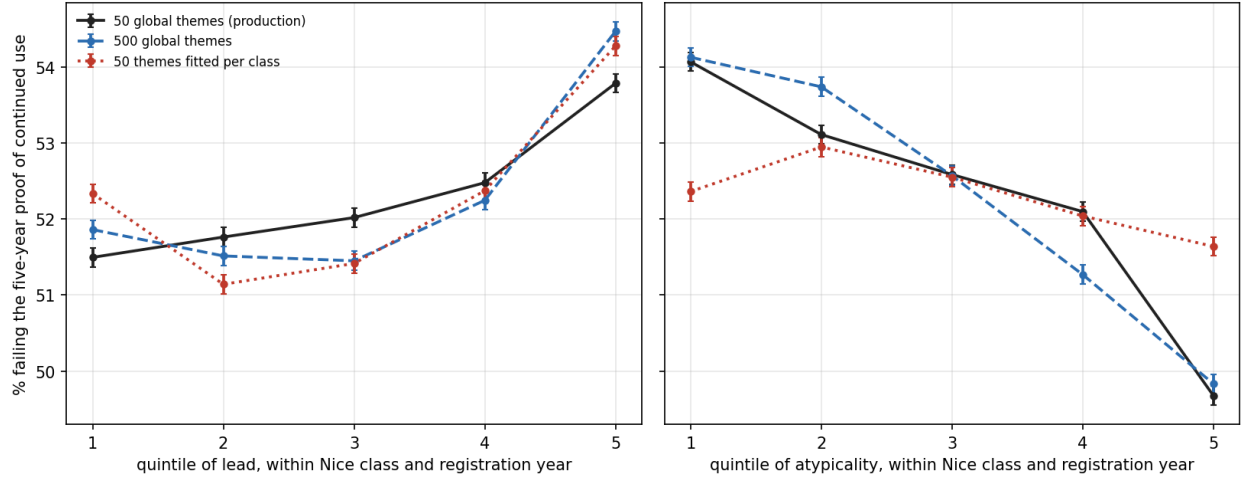


Figure 4: Failure at the five-year proof of continued use by quintile of lead (left) and of atypicality (right) under the three theme scorings, on the identical 3.10 million registrations 2002–2018; quintiles cut within Nice class and registration year, no controls. Whiskers are 95% binomial intervals per quintile, uncorrected for owner clustering. The lead gradient survives every scoring; the atypicality gradient does not survive per-class themes.

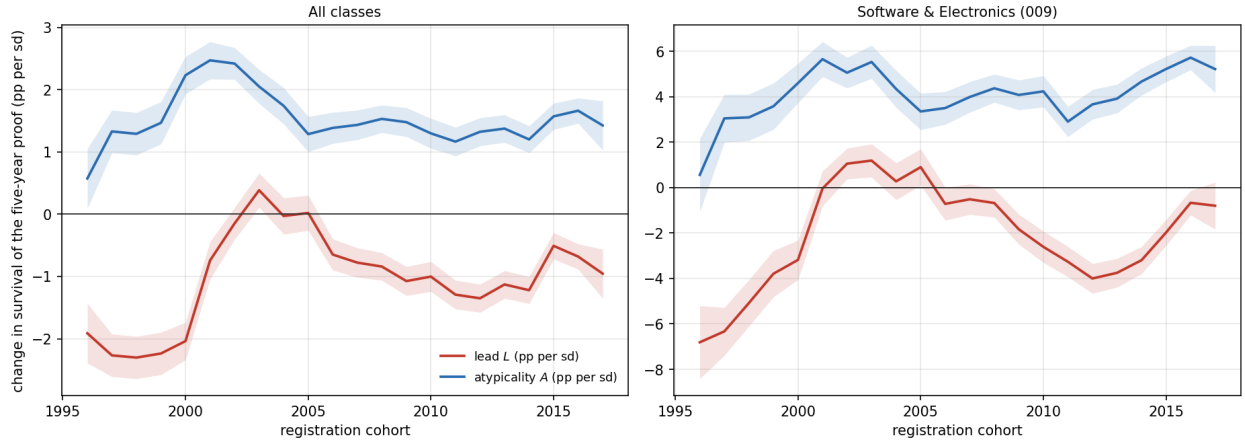


Figure 5: Survival of the five-year proof of continued use, per registration cohort, as continuous coefficients: the change in survival, in percentage points, for a one-standard-deviation increase in lead (red) and in atypicality (blue), entered jointly, both standardized within Nice class and cohort. Left: all classes pooled ($n = 3,150,442$ settled registrations); right: software and electronics alone. Bands are 95% intervals on OLS standard errors, uncorrected for the within-owner correlation of outcomes (0.38), so they understate the uncertainty somewhat.

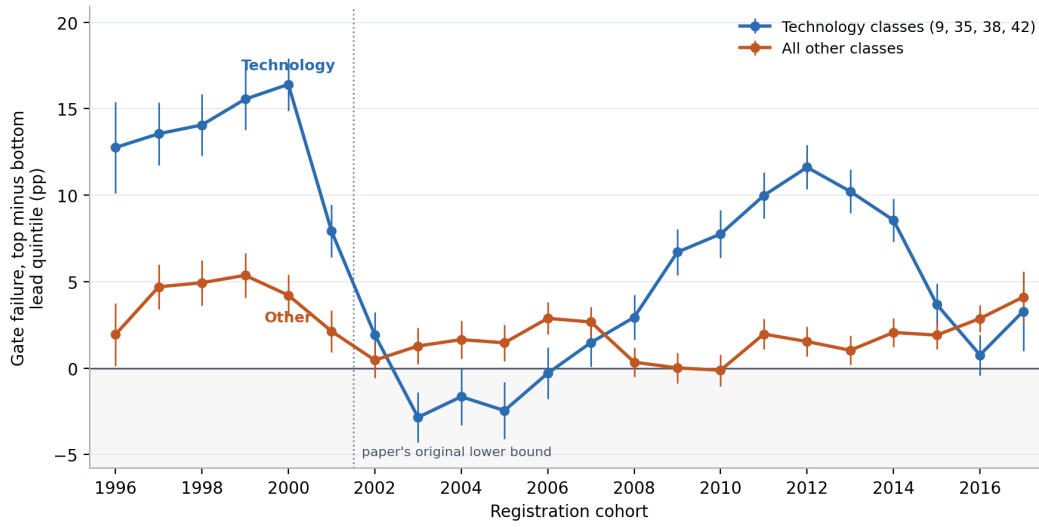


Figure 6: The leading penalty at the five-year proof by registration cohort, technology classes (009, 035, 038, 042) against all others. Each point is the failure rate of the most leading fifth of that cohort's registrations minus that of the most lagging fifth, lead quintiles cut within cohort and group, no controls, with 95% intervals, on settled registrations. Outside technology the penalty is small and almost flat across two decades; inside technology it swings from +16pp for the 2000 cohort to -3pp for 2003 and back to +12pp for 2012. The dotted line marks the 2002 cut that the cohort samples of Section 5 use.

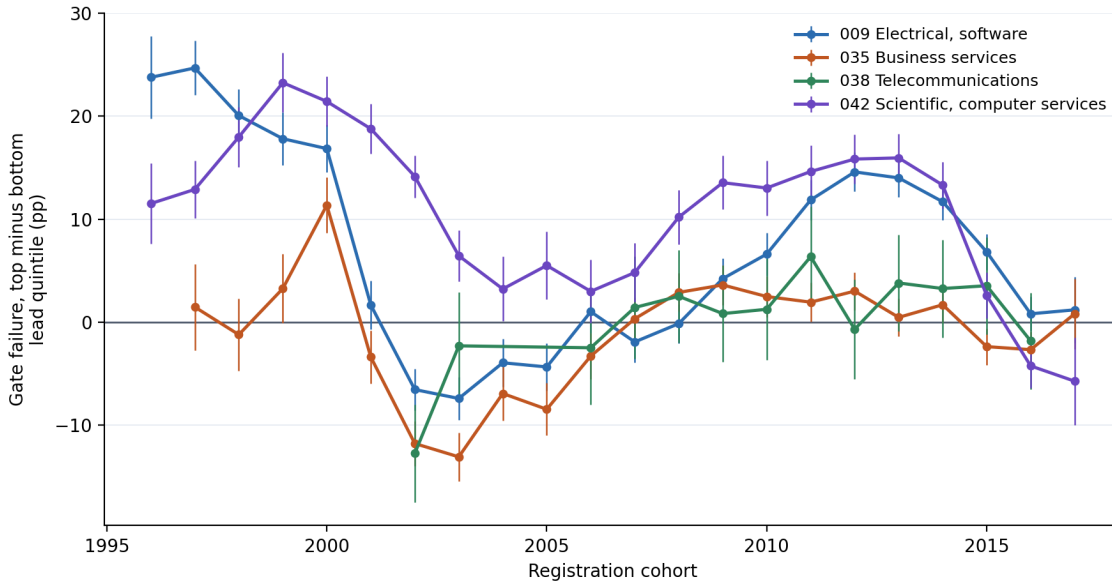


Figure 7: The leading penalty at the five-year proof by registration cohort, one line per technology class. Each point is the failure rate of the most leading fifth of that class and cohort's registrations minus that of the most lagging fifth — lead quintiles cut within class and cohort, no controls — with 95% intervals, on settled registrations. Three classes cross zero between 2001 and 2006; scientific and computer services does not.

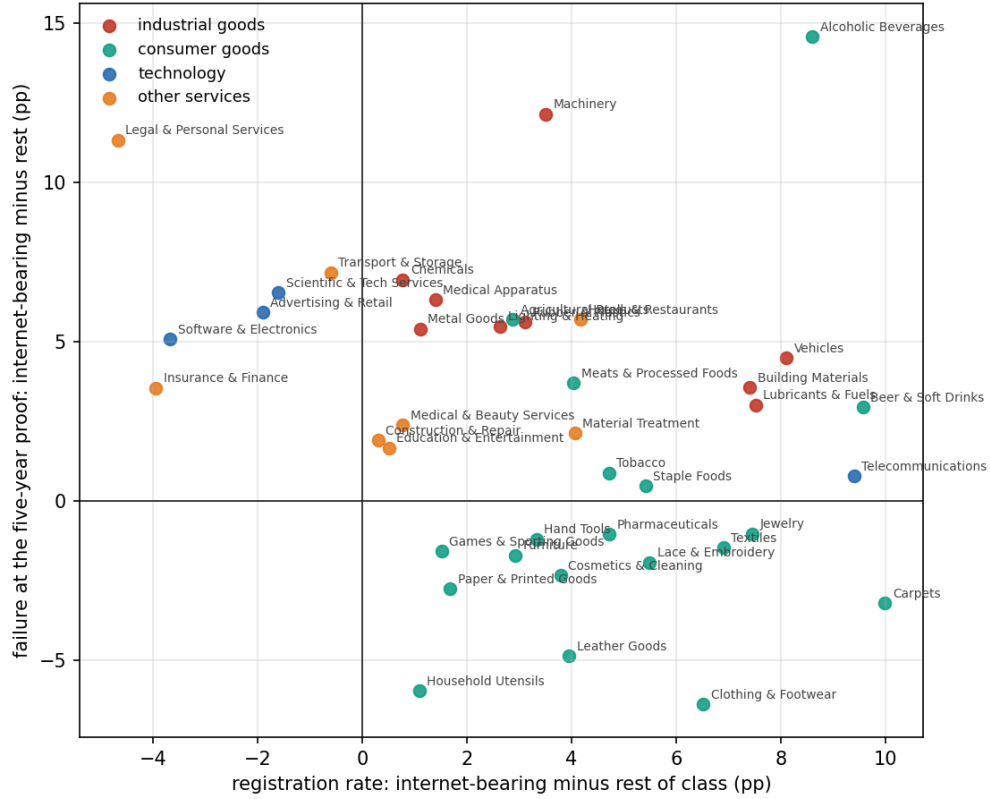


Figure 8: Each Nice class's internet-bearing filings against the rest of the class, on both steps at once. Horizontal: registration rate of internet-bearing class-records minus the rest of the class, filings 1995–2018. Vertical: failure at the five-year proof of continued use, internet-bearing minus the rest, registrations 2002–2018. Classes with fewer than 500 internet-bearing registrations are dropped. Classes in the upper-right quadrant grant internet filings more often and lose them more often.

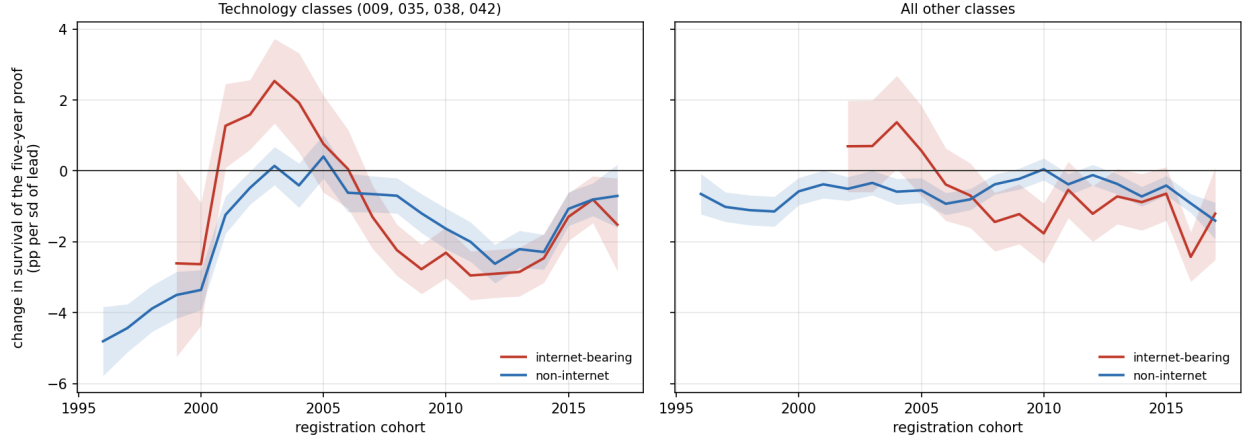


Figure 9: The per-cohort lead coefficient of Figure 5, split by the curated internet text pattern: change in survival of the five-year proof per standard deviation of lead, atypicality held, standardized within Nice class and cohort. Left: the four technology classes; right: all other classes. Bands are 95% intervals on OLS standard errors; cohort cells under 2,000 registrations are dropped, which is why the internet-bearing lines start late.

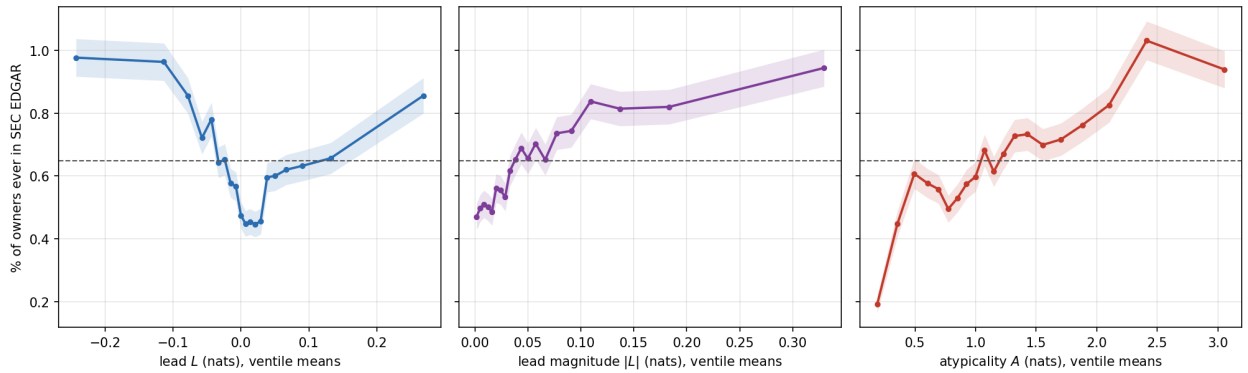


Figure 10: Share of registered debut owners (first registered filing 1995–2018, $n = 2,066,600$) ever appearing in SEC EDGAR, in twenty equal-count bins of debut lead L , lead magnitude $|L|$, and atypicality A , cut on the pooled distribution. Bands are 95% binomial intervals; the dashed line is the 0.65% base rate. These are raw rates carrying class, year and length composition; Table 9 gives the design-based estimates, under which the atypicality gradient is absorbed and the signed-lead U remains.

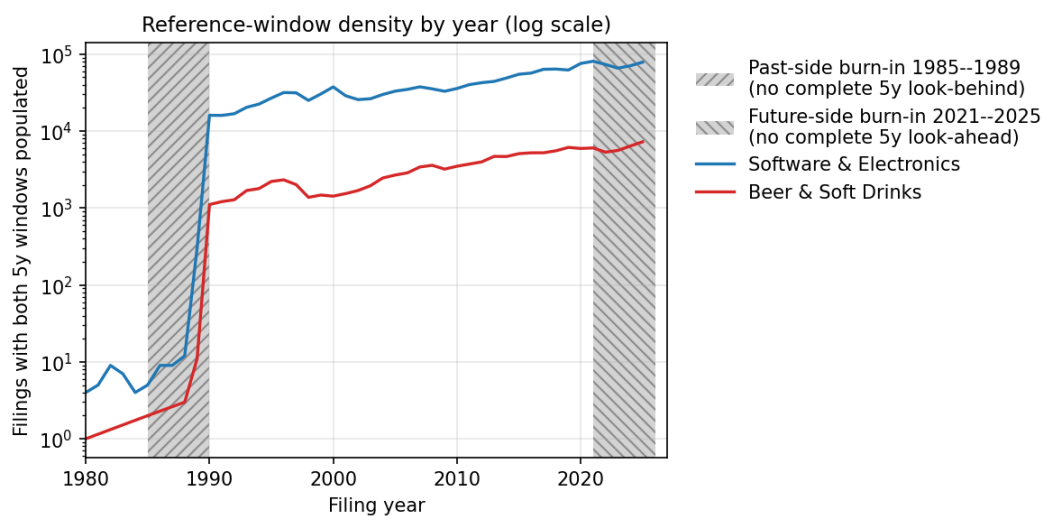


Figure 11: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the five-year window is incomplete; the dotted line marks the 1995 analysis cut.